

PROBLEM STATEMENT #31 | Retail, E-Commerce & Finance

Multi-Vendor Artisan E-Commerce Marketplace

1. Problem Context & Overview

An online marketplace enabling independent craftspeople to set up storefronts, manage product catalogs, receive orders, and receive automated split payouts with platform commission deductions.

Target Stakeholders / Actors: *Shopper, Artisan Vendor*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall split customer cart payments at checkout, allocating respective item earnings to multiple independent vendor accounts after deducting a 5% platform fee.

Sample Acceptance Criteria: Pass: Split payout calculations balance to total cart value; Fail: Payout calculation discrepancies.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The product catalog must support high-resolution image rendering with CDN caching delivering load times < 500 ms.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.